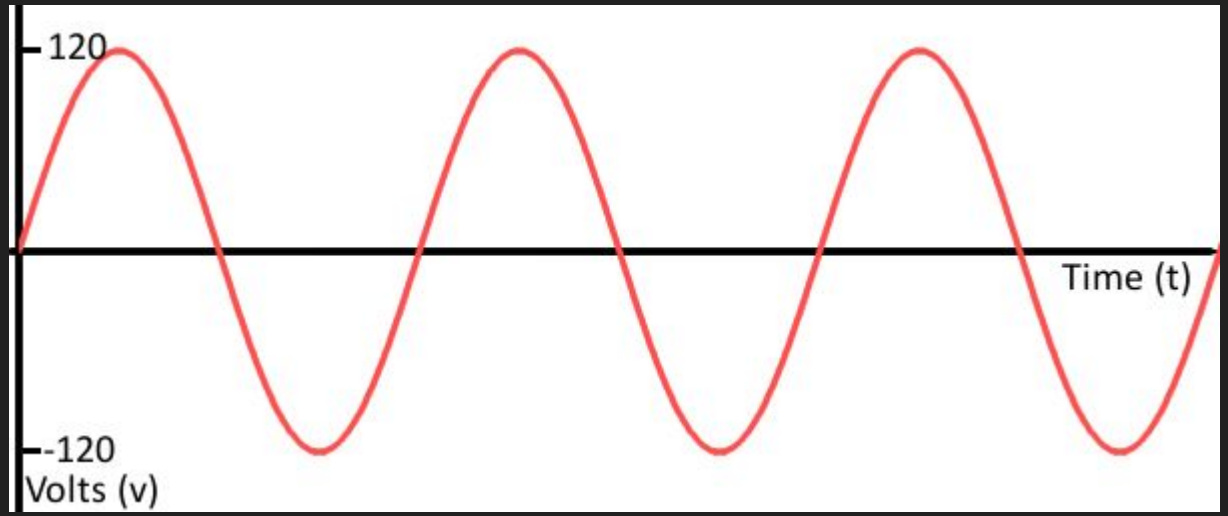
Arduino Analog Signals

Input and Output

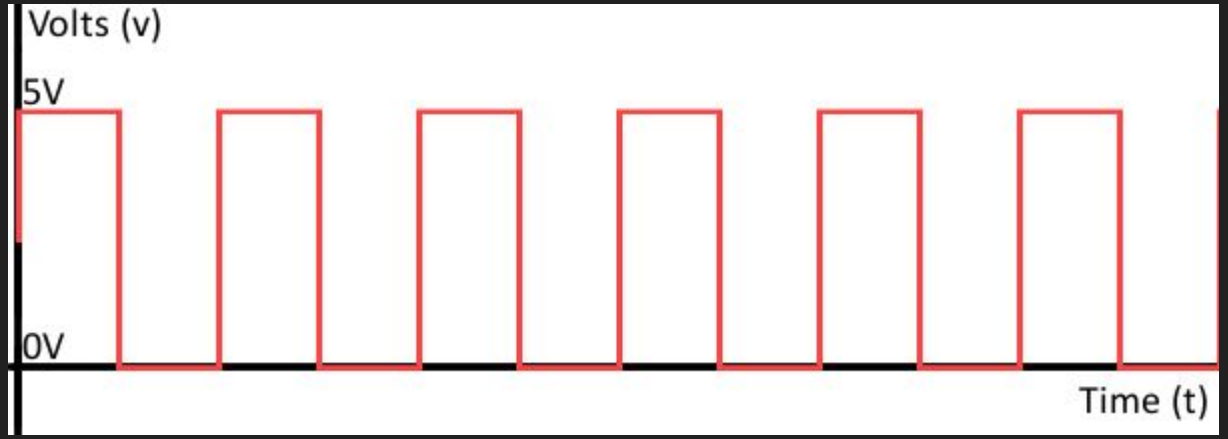
Analog Signals

Continually changing.



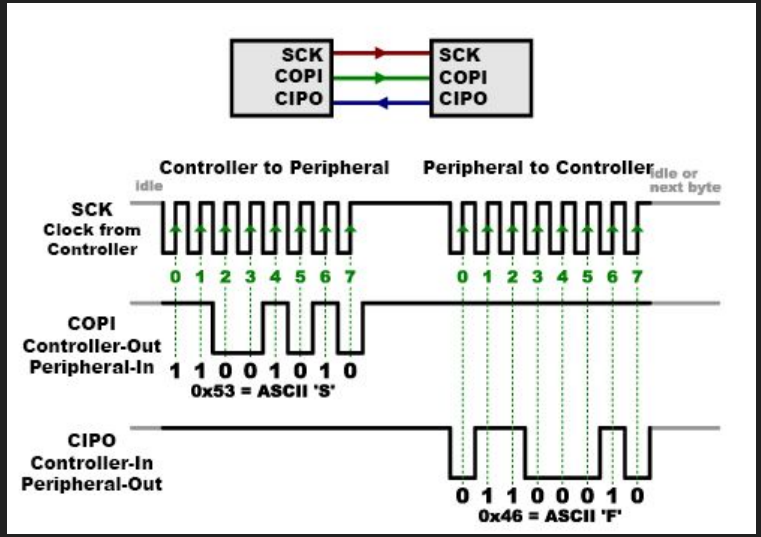
Digital Signals

Discrete values.



Digital Signal Communication

A clock determines the frequency when voltages are read and they are always either 0 or 1. The pattern of 0 and 1 determine the data.



Microcontrollers

Most are digital, but can work with analog signals by converting them to digital values. When analog signals are input by a microcontroller they use an ADC (analog to digital converter). When they want to output an analog signal they mimic this using PWM (pulse width modulation).

Arduino ADC

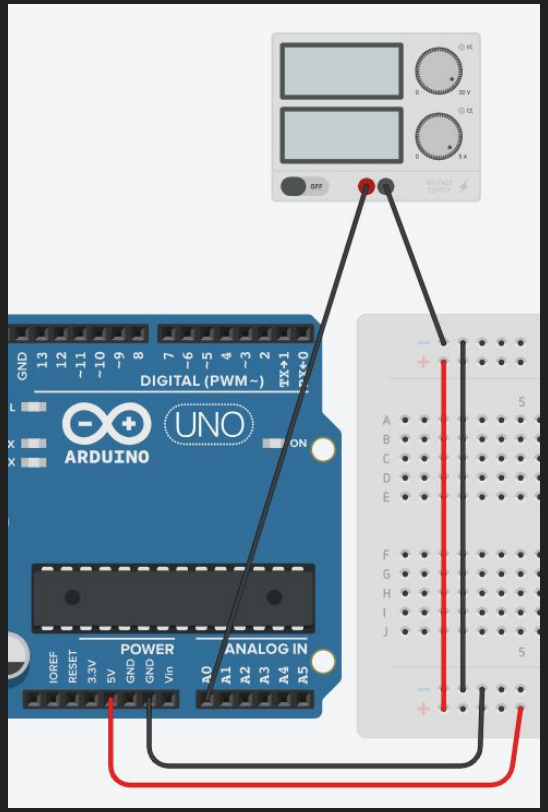
The Arduino has a set set of Analog Input pins using ADC. These pins take a varying voltage generated by an analog device/sensor and convert them to a digital value between 0 and 1023. These values are represented by 10 bits i.e.

0000000000 to 1111111111. This is why the Arduino ADC pins are referred to as 10 bit ADC.

To read the analog value we use the analogRead(analogPin) function.

The analog pins on the Uno are A0 to A5 and on the Mega they are A0 to A15.

Example: Design the following circuit/code




```
int analogValue=0;

void setup()
{
    Serial.begin(9600);
    pinMode(A0,INPUT);
}

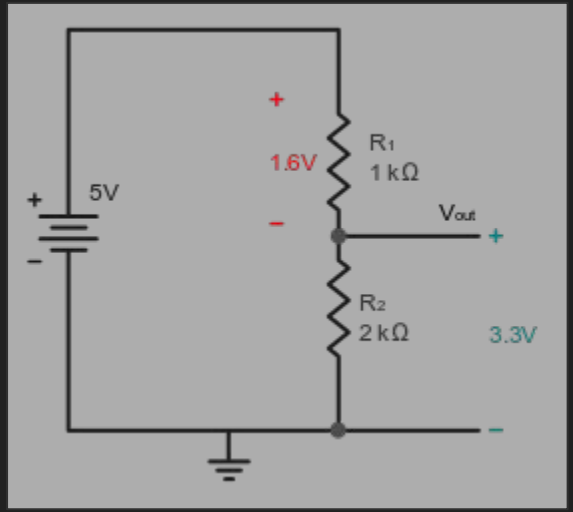
void loop()
{
  analogValue=analogRead(A0);
  Serial.println(analogValue);
}
```

What happens as the voltage is dialed from 0V to 5V?

What happens when the voltage exceeds 5V?

Generating Analog Voltages Using Analog Input Components

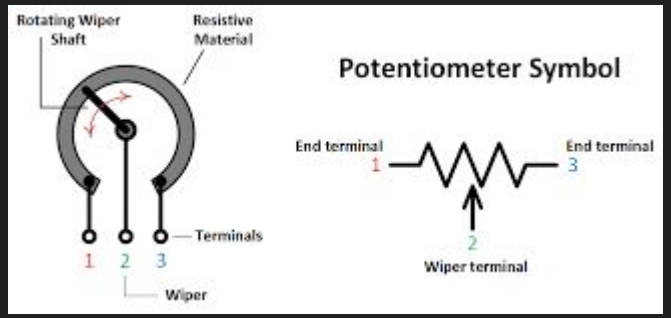
Most varying voltages are created using voltage dividers:



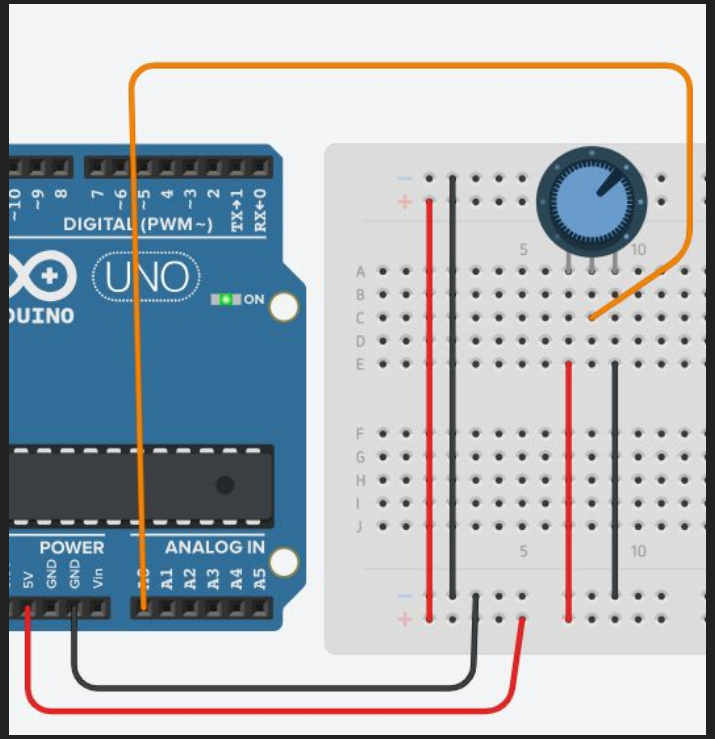
The voltage will drop 1.67V after the first resistor and another 3.33V after the second for a total of 5V. This leaves a voltage of 5-1.67=3.33V coming out of Vout. This can be measured using the ADC pins on the Arduino.

The Potentiometer

The POT is used to create a voltage that varies between two adjusting resistances.

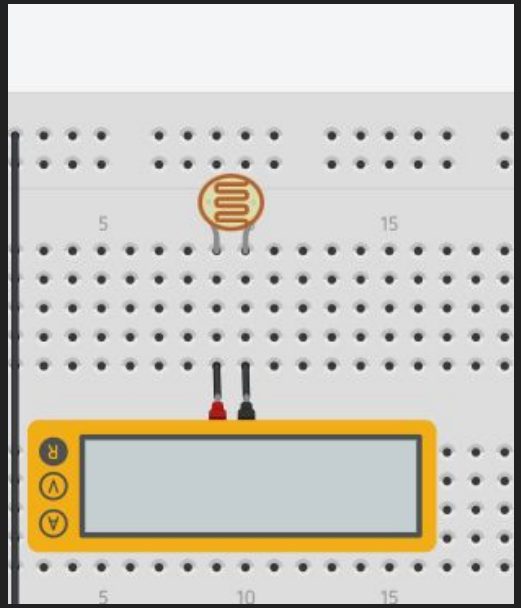


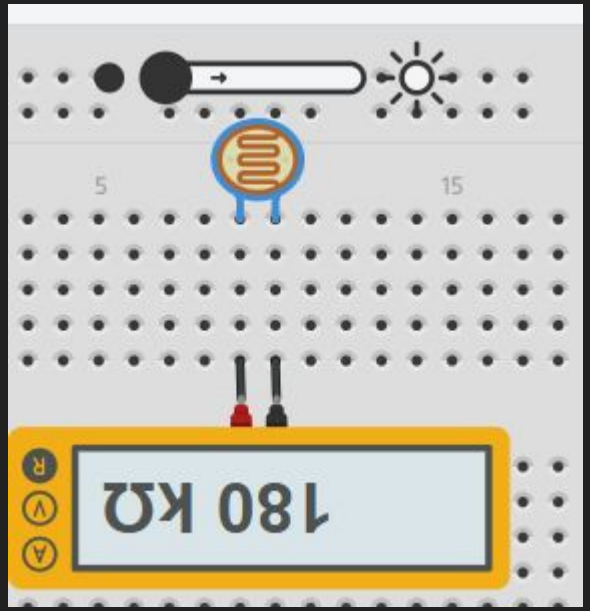
Try using the same code that was used with the power supply. What happens?

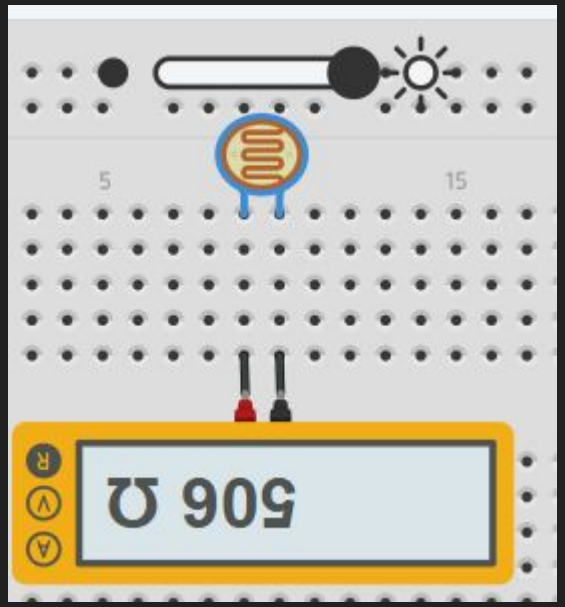


Photoresistor

Measure the resistance variance in a photoresistor (resistance changes depending on intensity of light).



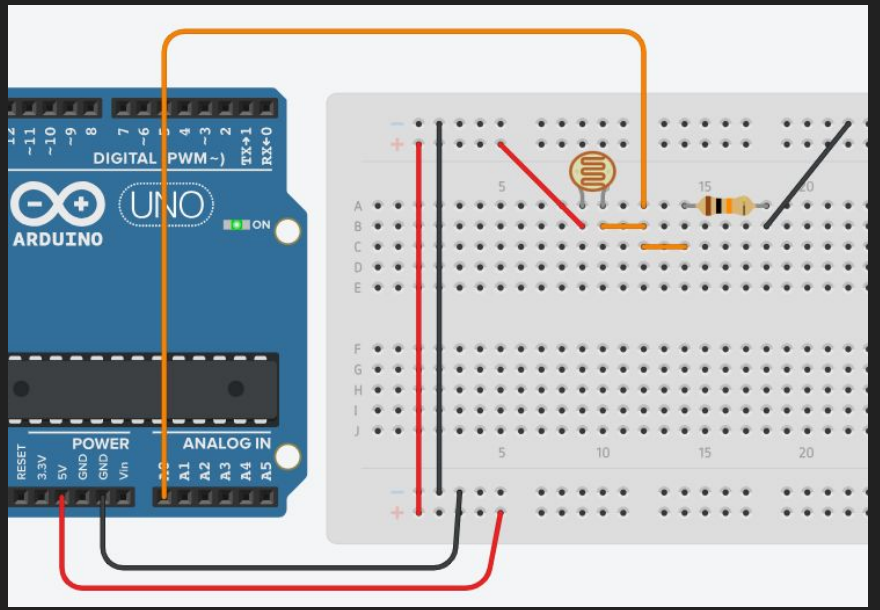




Create a Voltage Divider Using the Photoresistor

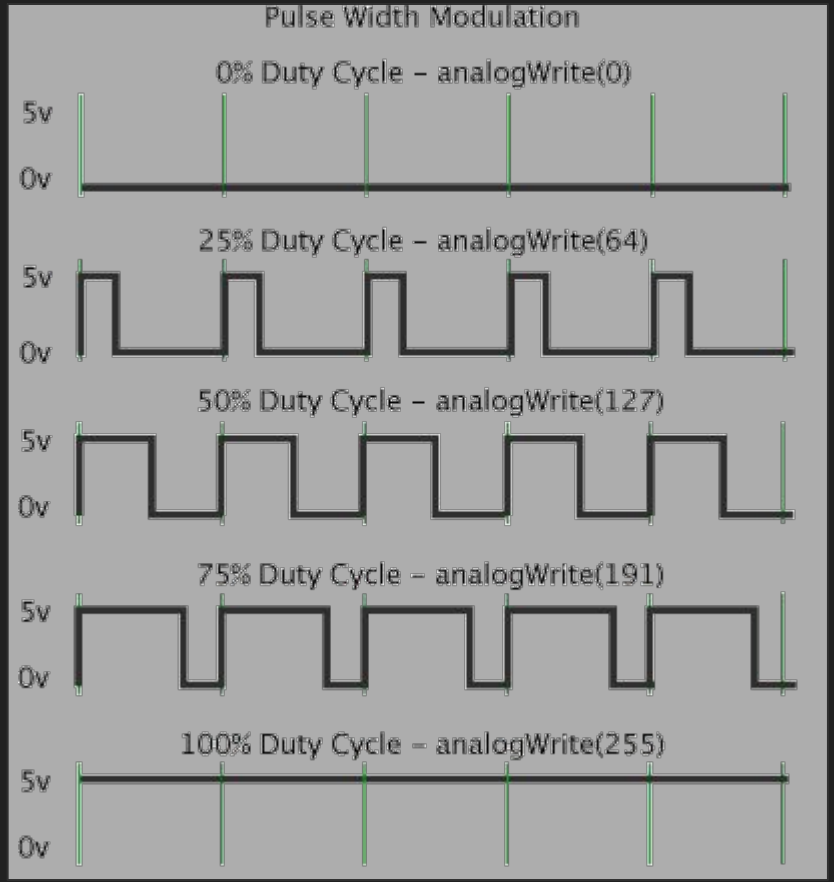
We can mimic the POT using the photoresistor and another resistor. The value of the second resistor will determine the range of voltages and therefore the range of values read via the analog input pin.

Run and test this circuit using the same code. It uses a 10 KOhm resistor. Try varying this and take note of the range of values.



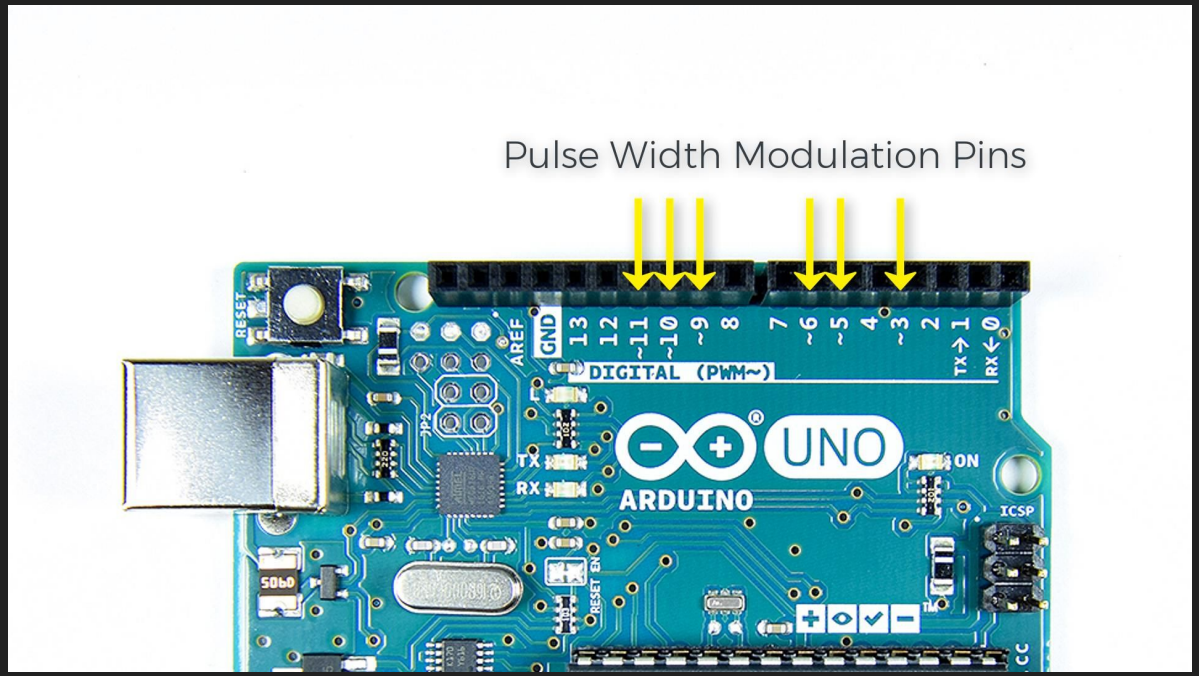
Analog Output

An analog output refers to a varying level of voltages. The Arduino is a fully digital device and can not actually output any voltage other than 0V or 5V. As a result we can only mimic a voltage in between using PWM(pulse width modulation). This is done by turning on and off a digital signal at varying frequencies. The frequencies are referred to as the Duty Cycle. The Duty Cycle is a percentage that refers to how much the signal is on relative to the clock signal duration. A 50% duty cycle means the signal is on for 50% of the clock signal time. We set this duty cycle using the analogWrite(duty cycle) function. The signal value uses 8 bits and therefore a number between 0 and 255. The higher the duty cycle, the longer the voltage is HIGH and the more it appears like the voltage is higher.



The PWM Pins

You can only use the digital pins labeled as PWM ~ for analog outputs:



Using PWM to Control the Brightness of an LED

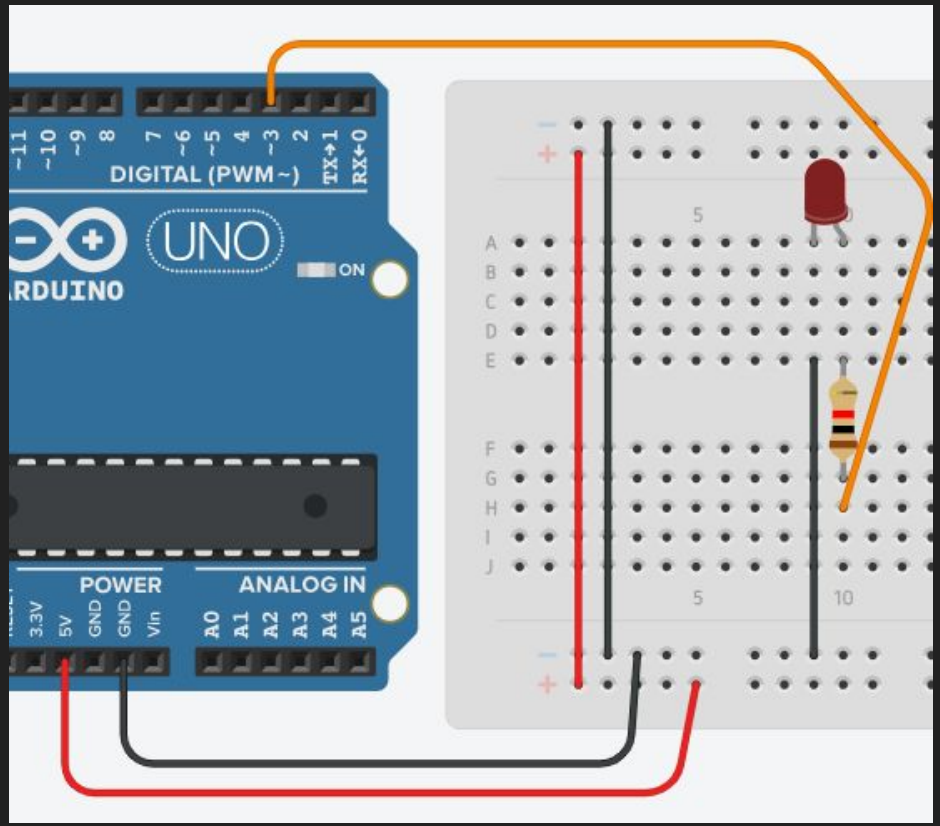
```
int analogValue=0;

void setup()
{
    Serial.begin(9600);
    pinMode(3,OUTPUT);
}

void loop()
{

  Serial.println(analogValue);
  analogWrite(3,analogValue);
  analogValue=analogValue+10;
  if(analogValue>255)
  {
   analogValue=0;
  }
  delay(100);

}
```



Questions and Exercises

1. Find and list the range of resistances for the following input components:

a. Photodiode 0Ω

b. Ambient Light Sensor764kΩ to 15.3kΩ

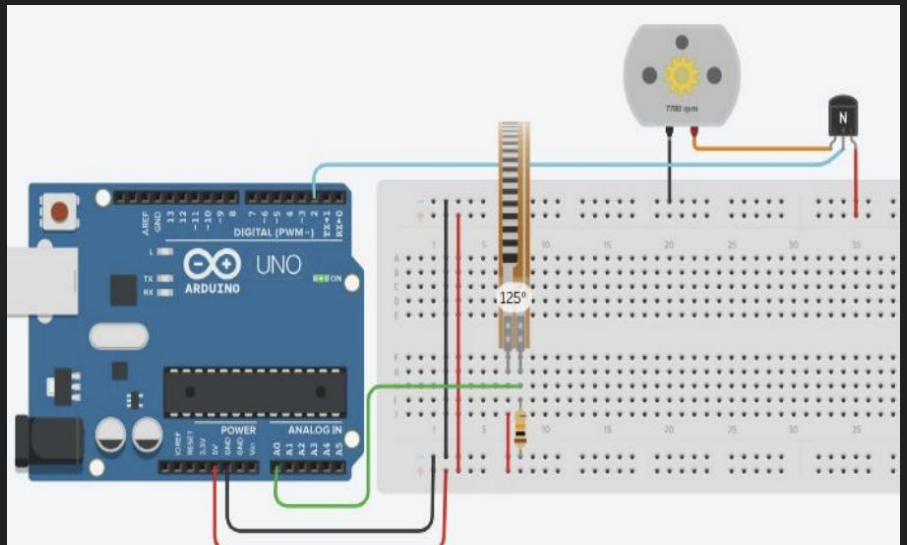
c. Flex Sensor **30kΩ to 163kΩ**

d. Force Sensor **1.2kΩ to 66.5kΩ**Tilt Sensor **10Ω to MΩ**

2. Create a circuit with two buttons, LED, piezo buzzer and DC motor (or hobby gear motor). As you press one button the LED brightens, the piezo buzzer gets louder and the DC motor trunks faster. As you press the other other button the LED dims, the piezo buzzer gets quieter and the DC motor slows down.

https://www.tinkercad.com/things/fAybcqWlqxc-fantabulous-bigery

3. Create a circuit with a hobby gear motor and a flex sensor. Change the speed of the motor as the flex sensor is bent. Research using the map function. Try and use this function to help you map the range of values the flex sensor generates to the range of values the PWM uses i.e. 0 to 255.



Assignment /20 TI

Create a circuit using the I2C LCD, a POT and a piezo buzzer. In the LCD create a moving character going left to right. Each time the character moves off the screen play a tone using a frequency of 2000. As the POT is dialed the speed of the moving character increases and the so does the frequency of the tone. You can use the following function to generate a tone:

tone(pin, frequency, duration)

Frequency should be between 2000 and 5000. The duration is in milliseconds. The pin is the digital pin attached to the positive side of the buzzer.

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